

Desigualdades

07/05/2016

$$\begin{array}{l|l} 2x+1 \geq 0 & 2x+1 > 0 \\ 2x = -1 & 2x > -1 \\ x = -\frac{1}{2} & x > -\frac{1}{2} \end{array}$$

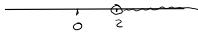


$$\begin{array}{l} -2x+1=0 \\ (-1) \cdot -2x = (-1) \cdot (-1) \\ 2x = 1 \\ x = \frac{1}{2} \end{array}$$

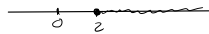
$$\begin{array}{l|l} -2x+1 > 0 & \\ (-1) \cdot -2x > (-1) \cdot (-1) & \\ 2x < 1 & \\ x < \frac{1}{2} & \end{array}$$

$$\begin{array}{l} -\infty < -4 < -2 < 0 < 2 < +\infty \\ 4 > 2 \\ (-1)4 > (-1)2 \\ -4 > -2 \\ -4 < -2 \end{array}$$

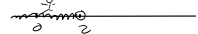
$x > 2 \Rightarrow x$ maior que 2



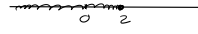
$x \geq 2 \Rightarrow x$ maior ou igual a 2



$x < 2 \Rightarrow x$ menor que 2



$x \leq 2 \Rightarrow x$ menor ou igual a 2



Sistema de Inequações

$$1-3x < 2x-24$$

$$-3x-2x < -24-1$$

$$(-1) \cdot -5x < (-1) \cdot -25$$

$$5x > 25$$

$$x > \frac{25}{5}$$

$$x > 5$$

$$a) \frac{x-1}{2} - \frac{1}{3}(4-2x) < \frac{5x-9}{6}$$

$$\frac{x}{2} - \frac{1}{2} - \frac{1}{3}(\frac{4}{3} - \frac{2x}{3}) < \frac{5x}{6} - \frac{3}{6}$$

$$\frac{x}{2} \cdot \frac{1}{2} - \frac{4}{6} + \frac{2x}{6} - \frac{5x}{6} + \frac{3}{6} < 0$$

Inequação < 0

m.m.c(2,3,6) = 6 $\Rightarrow$ Denominador

$$\frac{3x-3-8+4x-5x+9}{6} < 0$$

$$3x-3-8+4x-5x+9 < 0$$

$$2x-2 < 0$$

$$2x < 2$$

$$x < \frac{2}{2}$$

$$x < 1$$

$$\frac{6 \cdot \frac{x}{2} - \frac{4}{2} \cdot \frac{1}{3} + \frac{4 \cdot 2x}{3}}{6} < \frac{6 \cdot \frac{5x}{6} - \frac{6 \cdot 3}{6}}{6}$$

$$3x-3-8+4x-5x+9 < 5x-9$$

$$\frac{x}{5} \geq \frac{3}{8} \text{ Padião}$$

Inequação ≥ 0

$$\frac{x}{5} - \frac{3}{8} \geq 0$$

$$\text{m.m.c}(5,8) = 40$$

$$\frac{8x-15}{40} \geq 0$$

$$8x-15 \geq 0$$

$$8x \geq 15$$

$$x \geq \frac{15}{8}$$

$$40 \cdot \frac{x}{5} \geq \frac{3}{8} \cdot 40$$

$$\text{m.m.c}(5,8) = 40$$

$$8x \geq 15$$

$$x \geq \frac{15}{8}$$

$$6 \cdot \frac{x}{8} \geq \frac{3}{8} \cdot 5$$

$$x \geq \frac{15}{8}$$

$$3e.) \frac{2-x}{5} \geq \frac{4x+1}{3}$$

$$\frac{2-x}{5} - \frac{x}{5} \geq \frac{4x}{3} + \frac{1}{3}$$

Inequação ≥ 0

$$\frac{2}{15} - \frac{x}{15} - \frac{4x}{15} - \frac{1}{15} \geq 0$$

$$m.m.c(6,3) = 15$$

$$\frac{6-3x-20x-5}{15} \geq 0$$

$$6-3x-20x-5 \geq 0$$

$$-23x + 1 \geq 0$$

$$(-1) - 23x \geq -1 \quad (-1)$$

$$23x \leq 1$$

$$x \leq \frac{1}{23}$$

$$\frac{2-x}{5} \geq \frac{4x+1}{3}$$

$$15 \frac{2-x}{5} - \frac{x}{5} \geq 15 \frac{4x}{3} + \frac{1}{3}$$

$$m.m.c(5,3) = 15$$

$$6-3x \geq 20x+5$$

$$-3x-20x \geq 5-6$$

$$(-1) - 23x \geq -1 \quad (-1)$$

$$23x \leq 1$$

$$x \leq \frac{1}{23}$$

$$4. \frac{(2-x)}{5} \geq \frac{(4x+1) \cdot 3}{3}$$

$$3. (2-x) \geq \frac{(20x+5)}{3} \cdot 3$$

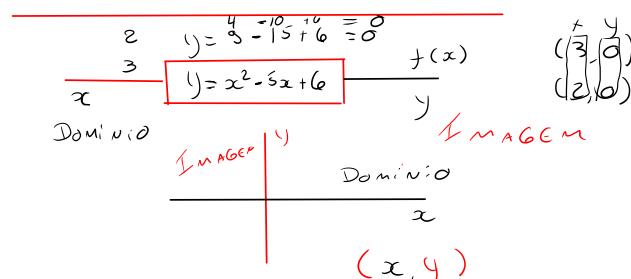
$$6-3x \geq 20x+5$$

$$-3x-20x \geq 5-6$$

$$(-1) - 23x \geq -1 \quad (-1)$$

$$23x \leq 1$$

$$x \leq \frac{1}{23}$$



$$0 = x^2 - 5x + 6$$

$$y = x^2 - 5x + 6 \Rightarrow \text{Parábola}$$

$$y = ax^2 + bx + c$$

$$\begin{cases} a = 1 \\ b = -5 \\ c = 6 \end{cases}$$

$$\Delta = b^2 - 4ac$$

$$\Delta = (-5)^2 - 4 \cdot 1 \cdot 6$$

$$\Delta = 25 - 24$$

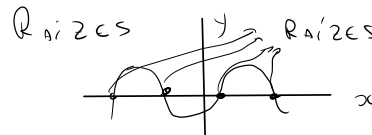
$$\Delta = 1$$

$$x = \frac{-b \pm \sqrt{\Delta}}{2a}$$

$$x = \frac{-(-5) \pm \sqrt{1}}{2}$$

$$x_1 = \frac{5+1}{2} = 3$$

$$x_2 = \frac{5-1}{2} = 2$$



Inequação de 2º Grau

$$y = x^2 - 5x + 6$$

$$x^2 - 5x + 6 > 0$$

$$y > 0$$

$$S = \{ x \in \mathbb{R} / x < 2 \text{ ou } x > 3 \}$$

$$y = x^2 - 5x + 6 < 0$$

$$y = x^2 - 5x + 6 \Rightarrow y < 0$$

$$2 < x < 3 \quad 2 \leq x \leq 3$$